

Question:
A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?
Answer Choices:
A. Epstein-Barr virus (EBV)
B. Measles virus
C. Human immunodeficiency virus (HIV)
D. Human papillomavirus

Reasoning + Answer:

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CD 2 1 is a receptor on B cells that plays a key role in B cell activation and humoral immune responses . It is part of the complement receptor complex and functions in enhancing B cell response to antigens by binding to complement proteins (like C 3 d) that are attached to pathogens .

The Epstein -Barr virus (EB V) is known to exploit B cells for its replication and immune evasion . EB V has been shown to interfere with the normal function of B cells and to potentially co-opt signaling pathways involving CD 2 1 . Importantly , EB V establishes latent infection in B cells , and the interaction between CD 2 1 and the viral components (such as EB V-encoded proteins) has been implicated in the immune response modulation and the pathogenesis of EB V-associated diseases like infectious mononucleosis and lymphomas .

Without CD 2 1 expression , B cells would be less responsive to complement signals that normally help in antigen recognition and B cell activation . Since EBV heavily relies on B cells for replication and immune evasion , impairing CD 2 1 signaling would likely disrupt the B cell response to EB V , directly affecting its pathogenesis .

In contrast :
- Measles virus primarily targets respiratory epithelial cells and macrophages , and its pathogenesis is less dependent on B cell CD 2 1 .
- HIV targets CD 4 + T cells , not B cells directly , and its pathogenesis is not reliant on CD 2 1 .
- Human papillomavirus (HPV) targets epithelial cells and causes cervical cancer ; its pathogenesis is unrelated to B cell signaling .

Thus , the genetic modification affecting CD 2 1 would most significantly impact the pathogenesis of Epstein-Barr virus .

< answer > Epstein-Barr virus (EB V)</answer >
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